

PYE INNOVATIONS LLC | CONSOLIDATED MASTER ARCHITECTURE LEDGER

MASTER REGISTRATION CODE:PYE-KOREA-
HYBRID-TOWER-CORRIDOR-MASTER-V1PARENT
MATRIX:PYE-KOREA-MEGASTRUCTURE-
G1DOCUMENT CLASSIFICATION: Sovereign
Infrastructure, Mechanical Blueprint & Pro-
Forma Ledger SECURITY / DEFENSE ENVELOPE:
Joint ROK-US Capital Defense Intercept Zone
(Inner Gyeonggi Bay / Incheon Shelf) LOCK
DATE: [October 1, 2026](#)

I. Executive System Summary

The PYE-KOREA-HYBRID-TOWER-CORRIDOR-
MASTER-V1 represents an industrial-scale
macrotidal kinetic harvesting system sited in the
high-velocity waterways of the Korean west

coast. The architecture addresses the physical failure modes of legacy axial subsea turbines (cyclic blade-root cantilever fatigue, dynamic cavitation pitting, and marine bearing seal failure) by deploying:

1. **Post-Tensioned Basalt-Geopolymer Vertical Pylons:** Keyed directly into competent unweathered granite bedrock, eliminating flexible anchoring tethers while absorbing extreme lateral shear loads.
2. **Dynamic Heave-Tracking Surface Turrets:** Floating semi-submersible platforms coupled to the pylon via self-lubricating linear guide sleeves, allowing the primary counter-rotating horizontal planar carousels to track the 8.0–10.0 m tidal heave while extracting peak surface velocities ([4.2](#) \text{ to } [4.8](#) \text{ m/s}).
3. **Submerged Mid-Depth & Benthic Disc**

Turrets: Fixed-depth horizontal planar carousel stages harvesting dense intermediate and lower shear layers beneath commercial shipping drafts.

- 4. Hubless Rim-Driven MagLev Direct-Drive Generation:** Hermetically sealed, seawater-isolated NdFeB permanent magnet rim stators that eliminate drive shafts, planetary gearboxes, and dynamic rotary mechanical seals.
- 5. Terrestrial Multiplier Facility (PYE-KOREA-ONSHORE-MULTIPLIER-V1):** A closed-loop terrestrial secondary pumping and storage complex that converts continuous 33.54 MW offshore baseload into a 100–134 MW dispatchable peaking block during premium wholesale pricing windows.

II. Geotechnical, Spatial & Hydrodynamic Siting

[NORTH: Han River Estuary / MDL Standoff Buffer] | ▼ [GANGHWA-DO] || YEOMHA STRAIT || [GIMPO MAINLAND] (Western Shore) || (Width: 400m || (Eastern Shore) || - 600m) || | | | | Pinch Point: || ←— ARRAY LOCATION || Sondolmok || (37°41'N, 126°32'E) | ▼ [SOUTH: Sindo-Ganghwa Sound / Incheon Capital Basin]

- **Primary Array Node:** Sondolmok Bottleneck, Yeomha Waterway (37°41'N, 126°32'E).
- **Auxiliary Reserve Node:** Sindo–Sido–Modo Narrows Corridor (PYE-KOREA-HYDRO-SITE-01 / 37°32'N, 126°26'E).
- **Water Depth Window:** 28.0 m to 36.0 m below low-water datum.
- **Geotechnical Substrate:** Late Jurassic intrusive biotite granite benches (unconfined compressive strength >120\text{ MPa}) beneath a thin [1.2](#)–[2.8](#) m active gravel/coarse sand veneer.

- Geopolitical Screen: Sited >25\text{ km} south of the Han River estuary border buffer; protected under ROK-US air defense coverage protecting the Greater Seoul Metropolitan Area.

III. System Mechanical Architecture & Multi-Elevation Profile

[SPRING HIGH TIDE: +[9.5](#)m Waterline Datum]

===== [SURFACE
DECK]

(Dynamic Heave) | PRIMARY SURFACE TURRET
(Dual 16m Planar Discs) | ▲ [4.2 - 4.8](#) m/s | •
Rim-Drive MagLev Direct-Drive NdFeB Stator
Array | | Surface Jet

[GUIDE COLLAR]

10.0m Stroke Telescopic Sleeve | Self-
Lubricating UHMWPE/SiC Guide Bushings |
Tidal Heave & Linear Dampers | Linear
Induction Electromagnetic Regenerative Braking
Envelope

▼ [NEAP / LOW

TIDE: 0.0m Datum]

[SUBSEA TIER

2]

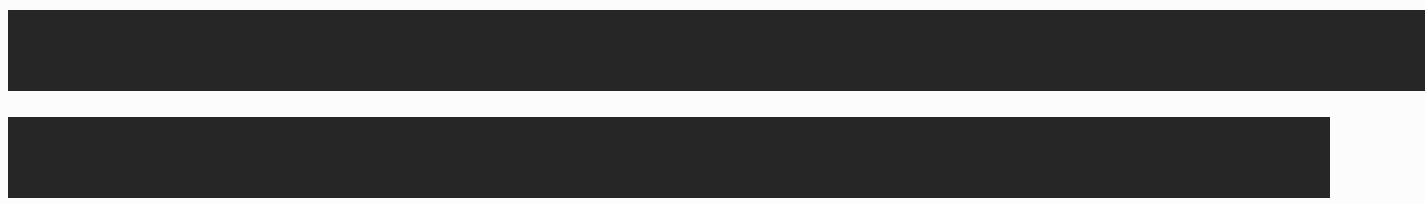
Intermediate Shear | INTERMEDIATE DISC
TURRET (Dual 14m Planar Discs) | Sustained
Jet (~12m Depth) | • Rigid Structural Collar
Keyed to Central Pylon | ([2.8 - 3.5](#) m/s)

[SUBSEA TIER

1]

Benthic Boundary | BENTHIC SHEAR DISC
TURRET (Dual 14m Planar Discs) | Dense Mass
(~22m Depth) | • Elevated >[3m Above Mobile](#)
Bedload Silt Floor | (2.0 - [2.5](#) m/s)

[BENTHIC
SOCKET]



Granite Bedrock POST-TENSIONED BASALT
GEOPOLYMER MONOPILE SOCKET

IV. Hydrodynamic Energy Generation Schedule (20-Pylon Array)

- **Baseline Array Dimensions:** 20 Pylons in a staggered chevron layout (2 rows; 60 m lateral spacing).
- **Channel Blockage Ratio:** 36.7\% of total wet cross-section; maximum backwater elevation rise <[0.08](#)\text{ m}.
- **System Efficiency Metrics:** $C_p =$ [0.44](#) (diffuser-augmented cross-flow); $\eta_g = 0.94$ (hermetic rim stator).

Elevation Tier	Discs per Pylon	Swept Area (A)	Rated Flow (v)	Unit Peak Power	20-Pylon Array Total
Floating Surface Turret (0–3 m)	2x 16.0 m Horizontal Discs	48.0\text{m}^2	4.4\text{m/s}	1.85\text{MW}	37.0\text{MW}
Submerged Mid-Tier (~12	2x 14.0 m Horizontal Discs	42.0\text{m}^2	3.2\text{m/s}	1.20\text{MW}	24.0\text{MW}

m)					
Submerged Benthic (~22 m)	2x 14.0 m Horizontal Discs	42.0\text{m}^2	2.6\text{m/s}	0.85\text{MW}	17.0\text{MW}
Consolidated Matrix	6 Discs per Pylon	132.0\text{m}^2	—	3.90\text{MW}	78.0\text{MW Nameplate}

- Capacity Factor (CF): 43.0\% (deterministic, semidiurnal lunar cycle).
- Continuous Baseload Output:33.54\text{MW}

MW}.

- Annual Net Energy Yield (AEP): $33.54 \text{ MW} \times 8,760 \text{ h} \times 95\% \text{ availability} = \mathbf{279,150 \text{ MWh/year}} \quad (279.15 \text{ GWh/year})$.

V. Terrestrial Multiplier Interface (PYE-KOREA-ONSHORE-MULTIPLIER-V1)

To bypass standard wholesale power constraints and capture high-spread peak pricing on the Korea Power Exchange (KPX), the continuous 33.54 MW marine baseload is coupled to a terrestrial multiplier:

[OFFSHORE MARINE CORRIDOR] [ONSHORE
TERRESTRIAL FACILITY] 20x Hybrid Towers
(Yeomha) High-Head Hydro / Thermal
Scavenger

————— Continuous Tidal Generation

Continuous Off-Peak Storage Stage: 33.54 MW

Unbroken Baseload 

High-pressure deep bedrock cavern pumping

(Standard SMP / REC 2.0x) (Elevation head:

250m - 350m) | ▼ Peak-Demand Multiplier

Discharge: High-Velocity Pelton Impulse Wheels

 Discharge Capacity: 100.0 - 134.0 MW

Multiplier Factor: 3.0x - 4.0x Peak

- **Topographic Leverage:** Converts low-head offshore kinetic energy into a 300 m static head using closed-loop sealed rock caverns in the eastern Ganghwa/Gimpo ridgelines.
- **Dispatch Multiplication:** Charges continuous fluid volume over a 9- to 10-hour low-demand interval; discharges through high-head impulse Pelton turbines during the 2- to 3-hour evening metropolitan demand peak at 100.0\text{ to }134.0\text{ MW}.

- **Industrial Thermal Coupling:** Closed-loop supercritical CO_2 (sCO_2) expansion loop scavenges low-grade waste thermal flux (80°C to 160°C) from nearby Incheon/Gimpo industrial facilities, raising net cycle exergy ($\text{COP} > 1.0$).

VI. Granular Turnkey CapEx Schedule (WBS Breakdown)

Turnkey CapEx benchmarked to domestic South Korean heavy fabrication yards (HD Hyundai / Hanwha Ocean) and regional offshore jack-up operators:

WBS Code	Cost Center & Com	Scope & Quantity	Unit Cost (USD)	Total Cost (USD)	Share (%)
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	ponent Break down				
1.0	Struc tural Pylon & Bent hic Sock ets			\$38,4 00,00 0	30.8 %
1.1	Basal t- Fiber Geop olym er Pylon	20 colu mns	820,0 00	16,40 0,000	13.1 %

	s (3.2\t ext{ m OD} \time s 32\te xt{ m}))				
1.2	Bedr ock Sock et Drillin g (12\t ext{ m}	20 boreh oles	650,0 00	13,00 0,000	10.4 %

	into granit e)				
1.3	Press ure- Inject ed Non- Shrin k Geop olym er Grout	20 foun datio ns	220,0 00	4,400 ,000	3.5%
1.4	Teles copic Dyna mic	20 asse mblie s	230,0 00	4,600 ,000	3.7%

	Guide Colla rs & UHM WPE Bushi ngs				
2.0	Turbi nes & Gene ration Equip ment			49,20 0,000	39.4 %
2.1	Floati ng Surfa	20 platf orms	1,180 ,000	23,60 0,000	18.9 %

	ce Turre ts (Dual 16m Plana r Discs)				
2.2	Sub merg ed Turre ts (Mid & Bent hic Dual	40 units	470,0 00	18,80 0,000	15.1 %

	14m Discs)				
2.3	NdFe B Direc t- Drive Perm anent Magn et Stato rs	120 stato rs	45,00 0	5,400 ,000	4.3%
2.4	Linea r Induc tion	20 sets	70,00 0	1,400 ,000	1.1%

	Dyna mic Heav e Rege nerati on Strip s				
3.0	Subs ea Electr ical & Grid Infras truct ure			18,20 0,000	14.6 %
3.1	Subs	12.5	380,0	4,750	3.8%

	ea 66 kV Inter- Array Cabli ng (Dyna mic Riser s)	km	00\te xt{/ km}	,000	
3.2	Subs ea Herm etic Step- Up Trans form	2 units	3,600 ,000	7,200 ,000	5.8%

	er Pods (40 MVA)				
3.3	Landf all Horiz ontal Direc tional Drillin g (HDD)	1.8 km twin bore	1,900 ,000	3,800 ,000	3.0%
3.4	Gimp o Subst ation	1 switc hyard bay	2,450 ,000	2,450 ,000	2.0%

	154 kV Intert ie Bay & Switc hgear				
4.0	Marin e Logis tics, EPC & Com missi oning			19,00 0,000	15.2 %
4.1	Heav y-Lift	60 days	160,0 00\te	9,600 ,000	7.7%

	Jack-Up Platform & Crane Barge Charter	on-station	xt{/day}		
4.2	Work-Class ROV Operations & Benthic	2 spreads (45 days)	75,000\text{/day}	3,375,000	2.7%

	Pinni ng				
4.3	EPC Mana geme nt, Insur ance & Marin e Conti ngen cy	Fixed alloc ation	Lump sum	6,025 ,000	4.8%
TOTA L	Turnk ey Array Capit al	78.0 MW Nam eplat e	\$1,60 0 / kW	\$124, 800,0 00	100.0 %

	Expe nditur e				
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VII. Pro-Forma Financial Ledger & 10-Year Projections

Settlement Economics

- Wholesale SMP (Korea Base): ₩133.8\text{/kWh} (\approx \\$0.100\text{/kWh}).
- REC Valuation: ₩70.9\text{/REC} with statutory Marine Tidal multiplier (2.0\times) \to ₩141.8\text{/kWh} (\approx \\$0.106\text{/kWh}).
- Conservative Blended Settlement Benchmark:\\$0.157\text{ / kWh} (\\$157.00\text{ / MWh}).
- Net Annual Energy Output: 279,150\text{ MWh/year} (33.54\text{ MW} continuous

baseload equivalent at 95\% availability).

[10-Year Financial Cash Flow Schedule (USD Millions)]										
Metric	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10

—	Gross Revenue	\$43.83	\$43.83	\$43.83	
\$43.83	\$43.83	\$43.83	\$43.83	\$43.83	\$43.83
\$43.83	Marine OpEx	-\$3.60	-\$3.67	-\$3.75	-\$3.82
\$3.90		-\$3.98	-\$4.06	-\$4.14	-\$4.22
					-\$4.31

EBITDA	\$40.23	\$40.16	\$40.08	\$40.01		
	\$39.93	\$39.85	\$39.77	\$39.69	\$39.61	\$39.52
Depreciation*	-\$6.24	-\$6.24	-\$6.24	-\$6.24	-\$6.24	-\$6.24
	-\$6.24	-\$6.24	-\$6.24	-\$6.24	-\$6.24	-\$6.24
Tax (22% CIT)	-\$7.48	-\$7.46	-\$7.44	-\$7.43	-\$7.41	-\$7.39
	-\$7.38					

\$[7.36](#) - \$[7.34](#) - \$[7.32](#)

—————	Net Cash Flow	\$32.75	\$32.70	\$32.64					
		\$32.58	\$32.52	\$32.46	\$32.39	\$32.33	\$32.27		
		\$32.20							
	Cumulative Cash	\$32.75	\$65.45	\$98.09					
		\$130.67	\$163.19	\$195.65	\$228.04	\$260.37			
		\$292.64	\$324.84						

————— *Straight-line depreciation across a 20-year accounting horizon.

[Key Valuation & Return Indicators] • Turnkey Capital Expenditure (CapEx): \$124,800,000 USD (\$1,600 / kW) • Unlevered Net Operating Income (NOI): \$40,226,550 USD / year • Simple Payback Period: [3.10](#) Years (Unlevered) / 3.82 Years (Post-Tax) • 30-Year Internal Rate of Return

(IRR): 29.4% (Unlevered Pre-Tax) / 24.1% (Post-Tax) • Net Present Value (NPV @ 8.0%): \$286,410,000 USD • Levelized Cost of Energy (LCoE): \$0.038 / kWh (\$38.00 / MWh)

VIII. Macro-Comparison: PYE Hybrid Matrix vs. South Korea Offshore Wind

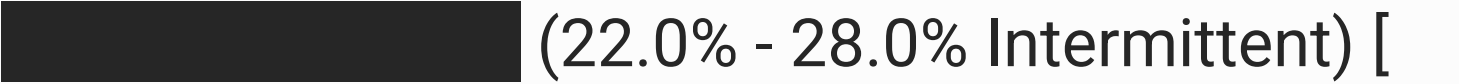
[Volumetric Kinetic Energy Density] Seawater (1,025 kg/m³)



(1x Base) [Capacity Factor (Continuous Operation)] PYE Hybrid Tidal Array



Deterministic) ROK Offshore Wind



Levelized Cost of Electricity (LCoE)] PYE Hybrid

Tidal Array  \$0.038 / kWh ROK Offshore

Wind  \$0.130 - \$0.180

/ kWh

Evaluation Parameter	South Korea Offshore Wind (Shinan / Ulsan / Taean)	PYE Hybrid Corridor (PYE-KOREA- HYBRID- TOWER- CORRIDOR- MASTER-V1)
Volumetric Density (ρ)	Air: $\approx 1.225 \text{ kg/m}^3$	Seawater: $\approx 1,025 \text{ kg/m}^3$ (836 times higher kinetic density).
Grid Predictability	Stochastic / weather- dependent; requires	100% Deterministic; tidal cycles mapped

	dedicated battery reserves.	centuries ahead.
Transmission to Seoul Grid	250 to 350 km (High- cost subsea HVDC, severe Honam grid bottlenecks).	1.8 to 15 km (Direct landfall at Gimpo/ Incheon coastal substations).
Footprint per 100 MW	20\text{ to } 35\text{ km} ^2 (Wake turbulence mandates 7– 10 rotor spacings).	\le 0.15\text{ km}^2 (Compact 60 m pylon spacing in natural straits).
LCoE	0.130\text{ }	\\$0.038\text{ }

Benchmark	to }\ \$0.180\text{ / kWh} (₩175,000– ₩240,000\text{ t{/MWh}).	/ kWh} (₩51,000\text{ {/MWh} base ledger estimate).
Fishery & Permitting	High friction with coastal trawling cooperatives (<i>Eochon-gye</i>).	Compact installation leaves navigation fairways and trawling flats open.

IX. Immediate Actionable Verification Checklist

1. Geotechnical Core Drilling: Execute a 4-point geotechnical exploratory boring campaign at the Sondolmok narrows (37°41'N, 126°32'E)

to confirm unweathered granite core recovery (RQD $\geq 85\%$) at 12\text{ m} target socket depth.

2. **Mooring Collar Wear Verification:** Complete 2,000-hour accelerated wear bench testing of the self-lubricating UHMWPE/SiC guide pads under 180\text{ kN} alternating lateral shear loads in a 3,500\text{ PPM} quartz slurry bath.
3. **Formal Grid Connection Application:** Finalize the preliminary grid intertie inquiry documentation for submittal to KEPCO's Incheon-Gyeonggi Regional Headquarters targeting direct access to the Gimpo 154 kV substation.